Python for Supply Chain

Session 4 · Thu 26 Nov 2026

Cleaning day

Today's question

What does the data look like when it is clean, and how much did we sell?

read_csv opens the file

IN DAILY LIFE

Opening the workbook.

IN EXCEL

File > Open.

THE LIMIT

The path must be exact.

```
import pandas as pd
folder = "/content/drive/MyDrive/Python for Supply Chain"
csv_file = "online_retail_ii_lesson.csv"
sales = pd.read_csv(f"{folder}/data/{csv_file}")
sales.head()
```

Look first: 105,335 rows and 8 columns

IN DAILY LIFE

First page, page count, cover.

IN EXCEL

Scrolling and COUNTA.

THE LIMIT

Looking first is part of the job.

```
sales.shape
sales.info()
sales.isna().sum()
```

A dict gives columns shorter names

IN DAILY LIFE

Relabelling the bins.

IN EXCEL

Editing the headers.

THE LIMIT

Uses the dict from session 2.

```
new_names = {
    "Invoice": "invoice", "Price": "price",
    "Quantity": "quantity", "Country": "country",
}
sales = sales.rename(columns=new_names)
```

A filter keeps only the rows you want

IN DAILY LIFE

Keeping only the good boxes.

IN EXCEL

Filter, then copy the result.

THE LIMIT

The original does not change. Use & and brackets.

```
is_germany = sales_clean["country"] == "Germany"
germany = sales_clean[is_germany]
germany.shape
```

Add a revenue column, then sort

IN DAILY LIFE

Writing a value on each box.

IN EXCEL

A formula column, then Sort Z to A.

THE LIMIT

No auto-update. Top row: 541431.

```
revenue = sales_clean["quantity"] * sales_clean["price"]
sales_clean["revenue"] = revenue
top = sales_clean.sort_values("revenue", ascending=False)
top.head(10)
```

A real date comes first, then the month

IN DAILY LIFE

A calendar column.

IN EXCEL

A date cell.

THE LIMIT

A real date first, then group by month.

```
dates = pd.to_datetime(sales_clean["invoice_date"])
sales_clean["invoice_date"] = dates
sales_clean["month"] = dates.dt.to_period("M")
```

Now in Colab

Notebook	s4_cleaning_day.ipynb
In class	Clean, count rows, write the log
At the end	Save sales_clean.csv to work

Clean data and 1,876,534.04 pounds

Recap in numbers

1. 105,335 raw rows, 8 columns
2. 101,458 clean rows, 10 columns
3. 1,876,534.04 pounds

Homework and recall

1. Re-run the notebook
2. Write the cleaning log
3. Try describe
4. Recall Fri 27 to Sun 29